KIRTI KOLARE

AI/ML Engineer · Backend Developer · Data Science

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PROFESSIONAL SUMMARY

AI/ML Engineer and backend developer with four internships across full-stack development, backend systems and applied machine learning. Builds and benchmarks LLM applications end-to-end, including cost-aware model routing, real-time voice agents, RAG hallucination guardrails and NL-to-SQL, using Python, FastAPI and modern GenAI tooling.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, SQL, C++
Back-end	FastAPI, Flask, Django, REST APIs, WebSockets, SQLAlchemy
AI / ML	LLM APIs (Claude, Gemini, Groq), LangChain, RAG, Prompt Engineering, Embeddings, scikit-learn
Data & Analytics	Pandas, NumPy, Power BI, Tableau, Excel
Databases	PostgreSQL, MySQL, MongoDB, Redis, Supabase, Vector Databases & Vector Search
Cloud & Tools	AWS, Docker, Git, GitHub, Streamlit, Figma, Claude Code, Codex

PROFESSIONAL EXPERIENCE

AI/ML Engineering Intern · **Wrrk.ai** *Jul 2026 – Oct 2026*

- Contributed to an all-in-one AI workspace unifying CRM, WhatsApp and voice/chat for business communication and automation.
- Built and maintained modules for analytics, unified inbox, email integration and residential proxy infrastructure.
- Guided new interns and potential clients through product demos, helping them understand and adopt the platform's features.

AI Intern · **Enzigma Solutions LLP** *Jan 2025 – Mar 2025*

- Built a semantic vector search pipeline on MongoDB using embeddings, replacing keyword matching for more relevant results.
- Redesigned LLM prompt templates across production use cases, improving output consistency and accuracy.
- Evaluated model outputs with engineers, iterated on prompts from real usage and presented findings to stakeholders.

Python Developer Intern · **JhaMobi Technologies Pvt. Ltd.** *Jun 2024 – Aug 2024*

- Built and deployed a full-stack book rental platform (Django, Bootstrap) end-to-end within an 8-week internship.

UI/UX Designer · **Marva AI LLP** *Jun 2023 – Aug 2023*

- Designed and shipped a food recommendation app, from Figma wireframes to a responsive Bootstrap/Django front-end.

PROJECTS

Cost-Aware Multi-Model LLM Router with Semantic Cache *FastAPI · Redis · Groq · Gemini · scikit-learn*

- Built a router that sends each query to the cheapest capable model, with a semantic cache catching near-duplicate queries.
- Cut cost ~23% vs. an expensive-model-only baseline with the lowest median latency (40-query eval, 100% classifier accuracy).

Interruptible Voice Agent for Support Ticket Triage *FastAPI · WebSocket · Silero VAD · faster-whisper · Groq · Piper TTS*

- Built a voice agent with true barge-in: callers can interrupt mid-response and it stops and responds instead of talking over them.
- Achieved 77 ms median cutoff and 2.2 s median recovery latency over 5 barge-in scenarios with real STT/LLM/TTS.

RAG Hallucination Guardrail *FastAPI · sentence-transformers · Cross-Encoder NLI · Groq*

- Built a verifier flagging RAG answers unsupported by retrieved context, comparing an NLI cross-encoder to an embedding baseline.
- Reached 0.725 F1 and 96.7% hallucination recall on a 60-entry eval set vs. 0.0 F1 with no guardrail, using untuned thresholds.

QueryPal: NL-to-SQL Platform with Multi-Tenant Isolation *Streamlit · LangChain · Groq/Llama 3.3 · Supabase*

- Two-agent LLM pipeline turning plain English into validated, self-correcting SQL, with live charts and per-account isolation.

CodeSense AI: Multi-Language Code Review Platform *scikit-learn · SMOTE · Gemini API*

- Random Forest defect-risk classifier (0.783 accuracy, 0.731 ROC-AUC) on 10,885 NASA JM1 modules, plus a Gemini code reviewer.

EDUCATION

M.Sc. Computer Science (CGPA 9.0/10) · **Indira University, School of Information Technology** *Jul 2025 – Jul 2027*

B.Sc. Data Science (CGPA 8.9/10) · **Symbiosis Skills & Professional University** *Aug 2022 – Aug 2025*

LEADERSHIP & ACTIVITIES

- Represented Indira University at Climate Corps Summer School 2026, Korea University, Seoul, in a 49-country cohort, and authored an AI & Energy research proposal.
- PG LinkedIn Learning Ambassador, School of Information Technology, Indira University (Present).